

PART-B

Name of the Student: Soumaraj Ghosh

Signature of the Student: Soumaraj Ghosh Roll No.: 230123061

Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	3	7	5	2	17

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

A-1 First canonical correlation r_1^2 is the sq. root of ^{leading} eigenvalue of $\Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1/2}$

So, r_1^2 is a leading eigen value of above matrix

as Y is 1×1 , $S_{yy}^{-1/2} = \frac{1}{\sqrt{\text{Var}(y)}}$

$$\text{So, } \frac{1}{\text{Var}(y)} S_{yx}^T S_{xx}^{-1} S_{yx}$$

$R^2 = \frac{SS_{\text{Reg}}}{SS_T}$

~~$$S_{yx} = \begin{bmatrix} \text{cov}(y, x_1) \\ \text{cov}(y, x_2) \\ \vdots \\ \text{cov}(y, x_q) \end{bmatrix}_{1 \times q}$$~~

$$Y = \begin{bmatrix} y \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_q \end{bmatrix}$$

$$S_{yx} = \begin{bmatrix} \text{cov}(y, x_1) \\ \text{cov}(y, x_2) \\ \vdots \\ \text{cov}(y, x_q) \end{bmatrix}_{1 \times q}$$

$$\text{Also } R^2 = \frac{SS_{\text{Reg}}}{SS_T}$$

$$SS_{\text{Reg}} = \sum (\hat{y}_i - \bar{y})^2$$

$$= \sum y_i^2 - 2 \sum y_i \bar{y} + n \bar{y}^2$$

$$SS_T = \sum (y_i - \bar{y})^2 = \sum y_i^2 - 2 \sum y_i \bar{y} + n \bar{y}^2 = \sum y_i^2 - n \bar{y}^2$$

Now by definition of variance, $SS_T = n \text{Var}(y)$

Here $n=1$ so, $SS_T = \text{Var}(y)$

So, ~~var~~ for q_i^2 we need eigen value of

$$\frac{1}{\text{var}(y)} S_{yx}^T S_{xx}^{-1} S_{yx}$$

So, lets find eigen value of $S_{yx}^T S_{xx}^{-1} S_{yx}$ & divide $\text{var}(y)$ at last

Lets check the dimensions

$$S_{yx}^T \rightarrow 1 \times q$$

$$S_{xx}^{-1} \rightarrow q \times q$$

$$S_{yx} \rightarrow q \times 1$$

So, $S_{yx}^T S_{xx}^{-1} S_{yx}$ will result in 1×1 matrix

$$\text{So, eigen value of } S_{yx}^T S_{xx}^{-1} S_{yx} = S_{yx}^T S_{xx}^{-1} S_{yx} \text{ as } 1 \times 1 \text{ matrix comes}$$

Now each variable is independent of one another in S_{xx}

so, $\text{cov}(x_i, x_j) = 0$ when $j \neq i$

So, S_{xx} is a diagonal matrix with entries as $\text{var}(x_i)$

i.e., $d_{ii} = \text{var}(x_i)$

$$\begin{aligned} \text{So, } S_{yx}^T S_{xx}^{-1} S_{yx} &= \begin{bmatrix} \text{cov}(\hat{y}, x_1) & \text{cov}(\hat{y}, x_2) & \dots \end{bmatrix} \begin{bmatrix} \frac{1}{\text{var}(x_1)} & 0 & \dots \\ 0 & \frac{1}{\text{var}(x_2)} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} \text{cov}(\hat{y}, x_1) \\ \text{cov}(\hat{y}, x_2) \\ \vdots \end{bmatrix} \\ &= \sum_{i=1}^q \frac{\text{cov}^2(\hat{y}, x_i)}{\text{var}(x_i)} \end{aligned}$$

Now, $\text{var}(x_i) = (x_i - \bar{x}_i)^2$ (By definition) $[\text{var}(x) = \mathbb{E}((x - \mathbb{E}(x))^2)]$

$$\text{Cov}(y, x_i) = (\hat{y} - \bar{y})(x_i - \bar{x}_i) \quad [\text{Cov} = \mathbb{E}((Y - \mathbb{E}(Y))(X - \mathbb{E}(X)))]$$

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix S . The eigenvalues of S are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are e_1, e_2, e_3 , and e_4 . You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{L}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{L}$ be $\tilde{l}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of S .

A-2 a) The loading matrix (estimated), $\tilde{L}$ is given by

(02)
$$\begin{bmatrix} \sqrt{4.2} e_3 & \sqrt{2.1} e_2 \end{bmatrix} = \begin{bmatrix} 2.0493 e_3 & 1.4491 e_2 \end{bmatrix}$$

as λ_3 and λ_2 are leading eigen values

b)
$$\tilde{h}_1^2 = l_{11}^2 + l_{12}^2$$

(01)
$$= 0.64 + 0.16$$

$$= 0.8$$

c) Let $e_3 = \begin{bmatrix} a_3 \\ b_3 \\ c_3 \\ d_3 \end{bmatrix}$

as e_3 is normalised

$$\sqrt{a_3^2 + b_3^2 + c_3^2 + d_3^2} = 1$$

Similarly $e_2 = \begin{bmatrix} a_2 \\ b_2 \\ c_2 \\ d_2 \end{bmatrix}$

and $\sqrt{a_2^2 + b_2^2 + c_2^2 + d_2^2} = 1$

Now $\tilde{h}_i^2 = l_{i1}^2 + l_{i2}^2$

$$\tilde{L} = \begin{bmatrix} \sqrt{4.2} a_3 & \sqrt{2.1} a_2 \\ \sqrt{4.2} b_3 & \sqrt{2.1} b_2 \\ \sqrt{4.2} c_3 & \sqrt{2.1} c_2 \\ \sqrt{4.2} d_3 & \sqrt{2.1} d_2 \end{bmatrix}$$

So, $\tilde{h}_1^2 = 4.2 a_3^2 + 2.1 a_2^2$

$\tilde{h}_2^2 = 4.2 b_3^2 + 2.1 b_2^2$

$\tilde{h}_3^2 = 4.2 c_3^2 + 2.1 c_2^2$

$\tilde{h}_4^2 = 4.2 d_3^2 + 2.1 d_2^2$

$$So, \sum \tilde{h}_i^2 = 4.2(a_3^2 + b_3^2 + c_3^2 + d_3^2) + 2.1(a_2^2 + b_2^2 + c_2^2 + d_2^2)$$

as eigenvectors are normalised, these 2 terms are 1

$$So, \sum \tilde{h}_i^2 = 4.2 + 2.1 \\ = 6.3$$

04

Sum of target eigenvalues of $S = 4.2 + 2.1 = 6.3$

—Hence, proved—

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

A-3 • We know that if $\sqrt{n}(X_n - \theta) \sim N(0, \sigma^2)$

$$\text{Then } \sqrt{n}(g(X_n) - g(\theta)) \sim N(0, \sigma^2 [g'(\theta)]^2)$$

Each X_i has mean μ & variance μ^2

So, $\sum X_i$ has mean $n\mu$ and variance $n\mu^2$

$\Rightarrow \frac{\sum X_i}{n}$ has mean μ & variance $\frac{n\mu^2}{n^2} = \frac{\mu^2}{n}$

$$\text{So, } \sqrt{n}(\bar{X}_n - \mu) \sim N(0, \frac{\mu^2}{n}) \quad \boxed{\text{(By CLT)}}$$

$$\text{So, } \sqrt{n}(g(\bar{X}_n) - g(\mu)) \sim N(0, \frac{\mu^2}{n} [g'(\mu)]^2)$$

$\hookrightarrow$ Asymptotic Variance

So, we need $\frac{\mu^2}{n} [g'(\mu)]^2$ is not depend on μ

$$\text{So, } \frac{\mu^2}{n} [g'(\mu)]^2 = c$$

$$\Rightarrow g'(\mu) = \sqrt{\frac{cn}{\mu^2}} = \frac{\sqrt{cn}}{\mu} = \frac{K}{\mu}$$

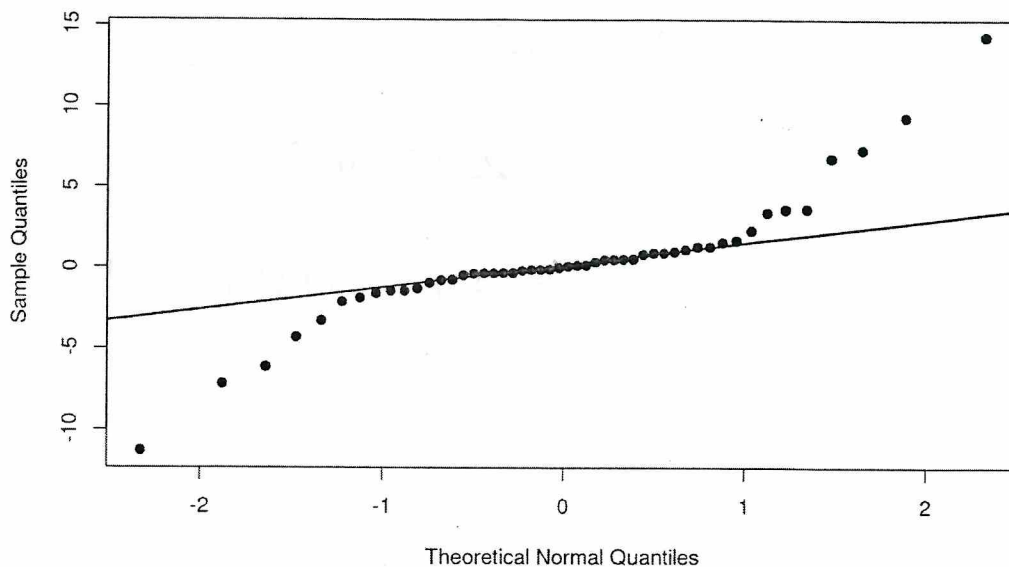
$$\text{So, } g(\mu) = \int \frac{K}{\mu} d\mu = K \ln \mu + C$$

OS

$$\text{So, } g(\bar{X}_n) = K \ln \bar{X}_n + C$$

$\hookrightarrow$ This is a variance stabilizing transformation.

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



The data given on which the Q-Q plot is drawn
is not normally distributed

The theoretical and normal quantiles follow ~~heavy tailed~~
~~almost~~ linear ~~scale~~ near 0 but near the
tails the sample quantiles take higher value. So
the distribution given is a heavy-tailed distribution
which deviates from normal.

02

A-1 Contd.

$$\begin{aligned}
 \text{So, } S_{yx}^T S_{xx}^{-1} S_{yx} &= \sum_{i=1}^n \frac{(\hat{y}_i - \bar{y})^2 (\cancel{x_i - \bar{x}_i})^2}{(\cancel{x_i - \bar{x}_i})^2} \\
 &= \sum (\hat{y}_i - \bar{y})^2 \\
 &= SS_{\text{Reg}} \text{ (Defined Before)}
 \end{aligned}$$

Also $SS_T = \text{Var}(y)$ (Defined before)

$$\begin{aligned}
 \text{So, eigenvalue} &= \frac{\sum (\hat{y}_i - \bar{y})^2}{\text{Var}(y)} \\
 &= \frac{SS_{\text{Reg}}}{SS_T} \text{ (By above 2 lines)} \\
 &= R_i^2
 \end{aligned}$$

$$\text{So, } R_i^2 = R^2 = \frac{SS_{\text{Reg}}}{SS_T}$$

— Hence, proved —

